CS2222 — Advanced Programming Techniques (deep C)

Tier	B
Phase (order)	2
Do this first	CS1111, CS1234
Estimated effort	6-8 weeks
Unlocks	CS2300 labs, CS3500, CS3300

Roadmap

1. Pointers and pointer arithmetic, dynamic memory, debugging with gdb/valgrind.
2. Bit manipulation, variadic functions, command-line args, file and directory I/O.
3. Macros, separate compilation, makefiles; inline assembly basics.

Video courses (all links checked on YouTube on 2026-10-22)

🎓 Main course: follow it start to end

- CS50x 2025 Lectures (13 videos). CS50x 2025: lecture 4 (Memory) and 5 (Data Structures) are the best pointer/malloc intro.

🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- Beginner C Videos (46 videos). Jacob Sorber *Beginner C Videos*: pointers, memory, make, the details.
- Debugging C Programs (38 videos). Jacob Sorber *Debugging C Programs*: gdb, valgrind, sanitizers.
- Unix Processes in C (27 videos). CodeVault *Unix Processes in C*: fork/exec/pipes (needed later in OS).

⚡ Short videos for intuition and revision

- Unix Threads in C (29 videos). CodeVault *Unix Threads in C*.
- C Programming and Assembly Language (32 videos). NPTEL-NOC IITM *C Programming and Assembly Language*: how C maps to assembly.

Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

Week	Lectures	What to do
1	CS50 lec 4-5	Memory, pointers, malloc, linked lists, hash tables. Do CS50 problem sets *Recover* and *Speller*.

Week	Lectures	What to do
2-3	Jacob Sorber Beginner C (alt)	Pointers to pointers, function pointers, const, bit fields, variadic functions, the preprocessor.
4	Jacob Sorber Debugging C (alt)	gdb, valgrind, AddressSanitizer, core dumps. Fix 5 memory bugs on purpose.
5-6	CodeVault Unix Processes + Threads (alt)	fork/exec/wait/pipe, pthreads, mutexes. Write a mini shell supporting pipes.
7	K&R ch. 5-8	Write your own malloc (free-list allocator) and a generic hash map with void*.

▼ **All lectures in *CS50x 2025 Lectures* (13 videos)**

1. CS50x 2025 - Introduction
2. CS50x 2025 - Lecture 0 - Scratch
3. CS50x 2025 - Lecture 1 - C
4. CS50x 2025 - Lecture 2 - Arrays
5. CS50x 2025 - Lecture 3 - Algorithms
6. CS50x 2025 - Lecture 4 - Memory
7. CS50x 2025 - Lecture 5 - Data Structures
8. CS50x 2025 - Lecture 6 - Python
9. CS50x 2025 - Artificial Intelligence
10. CS50x 2025 - Lecture 7 - SQL
11. CS50x 2025 - Lecture 8 - HTML, CSS, JavaScript
12. CS50x 2025 - Lecture 9 - Flask
13. CS50x 2025 - Lecture 10 - The End

Books, notes and websites

- K&R, *The C Programming Language* (the course's own text)
- Kernighan & Pike, *The Practice of Programming*
- *Modern C* (Jens Gustedt, free PDF)
- Beej's Guide to C

Practice / proof of understanding

Write your own `malloc` wrapper with leak tracking, a bit-vector library, and a tiny `grep` clone.

Note

Optional, but excellent preparation for OS and compilers.

What your IITM course sheet says

From the PDF: CS2222

Official description: - To be comprehensive in an imperative programming language - To learn more and deeper problem solving - To improve coding confidence

Official syllabus (trimmed):

> Overview of an Imperative Programming Language [1 week] Topics from the Programming Language such as [8 weeks] - pointers and pointer arithmetic, memory management, debugging - byte and bit manipulations, bit fields, string tokenization - variable number of arguments, command-line arguments, environment variables - file / directory operations, records, profiling, time utilities - macros, user-defined header files, separate compilation, makefiles - writing assembly code, exception handling, assertions - optimizing code for time and memory usage Library Usage and Problem Solving related to Various Domains such as [4 weeks] - numerical computing, crypto functions (math library, complex numbers, GNU scientific library) - string manipulations (string library) - geometry - matrix operations, linear algebra (random numbers) - logical thinking, puzzles Developing an Application [1--2 weeks] - putting the above learning together.

Official textbooks: Kernighan and Ritchie, "The C Programming Language", (2nd ed.) Prentice-Hall, 1988

Official references: C: How to program, H. M. Deitel, P. J. Deitel, 7th edition, Pearson Education, 2010. The Practice of Programming, Brian Kernighan and Robert Pike, Addison-Wesley Inc., 1999.